

Task	Prompt	Answer
adamic adar index	The task is to determine the Adamic-Adar index of two nodes in a graph. The Adamic-Adar index is the sum of the inverse logarithm of the degrees of the common neighbors of the two nodes. The input graph is guaranteed to be undirected. Here is an undirected graph containing nodes from 1 to 9. The edges are: (1, 5), (1, 4), (1, 8), (1, 2), (1, 3), (1, 7), (5, 2), (5, 3), (5, 4), (5, 9), (5, 6), (4, 8), (4, 9), (4, 7), (8, 2), (8, 3), (8, 6), (8, 7), (8, 9), (2, 3), (2, 7), (2, 6), (3, 9), (3, 7), (7, 6), (7, 9). Question: What is the Adamic-Adar index between node 4 and node 6? You need to format your answer as a float number.	1.5859
avg neighbor degree	The task is to determine the average degree of the neighbors of a node in the graph. Here is an undirected graph containing nodes from 1 to 8. The edges are: (1, 7), (1, 8), (1, 4), (7, 8), (8, 5), (2, 3), (2, 6), (3, 5). Question: What is the average neighbor degree of node 2 in the graph? You need to format your answer as a float number.	1.5
barycenter	The task is to determine the barycenter of a graph. The barycenter of a graph is also called the median. It includes the node that minimizes the sum of shortest path lengths to all other nodes. The input graph is guaranteed to be connected. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 2), (1, 6), (1, 5), (1, 7), (1, 4), (2, 6), (2, 5), (2, 7), (2, 4), (6, 4), (6, 5), (6, 7), (7, 3), (7, 4). Question: What is the barycenter of the graph? You need to format your answer as a list of nodes in ascending order, e.g., [node-1, node-2, ..., node-n].	[1, 2, 6, 7]
betweenness centrality	The task is to determine the betweenness centrality of a node in the graph. Betweenness centrality of a node u^* is the sum of the fraction of all-pairs shortest paths that pass through u^*. Here is an undirected graph containing nodes from 1 to 9. The edges are: (1, 6), (1, 4), (1, 8), (1, 9), (6, 2), (6, 7), (4, 7), (4, 5), (8, 3), (8, 5), (8, 7), (9, 3), (9, 5), (2, 7). Question: What is the betweenness centrality of node 5 in the graph? You need to format your answer as a float number.	0.0679
bfs	The task is to determine the breadth-first search (BFS) traversal order given a starting node. Stop when the BFS cannot be continued. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 2), (1, 5), (2, 3), (2, 4), (5, 3), (5, 4), (3, 4), (4, 7), (7, 6). Question: What is the breadth-first search (BFS) traversal order for the starting node 1? You need to format your answer as a list of edges, e.g., [(u1, v1), (u2, v2), ..., (un, vn)].	[(1, 2), (1, 5), (2, 3), (2, 4), (4, 7), (7, 6)]